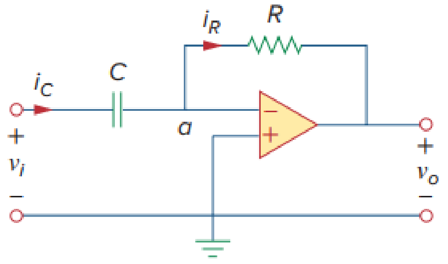
# Chapter 6, Problem 14PP

## Problem

The differentiator in Fig. 6.37 has $R$ = 100 kΩ and $C$ = 0.1 $\mu$F. Given that $vi$ = 5$t$ V, determine the output $vo$.

**Fig. 6.37:** An op amp differentiator.




## Step-by-step solution

**Step 1** of 2

Consider the following differentiator circuit:

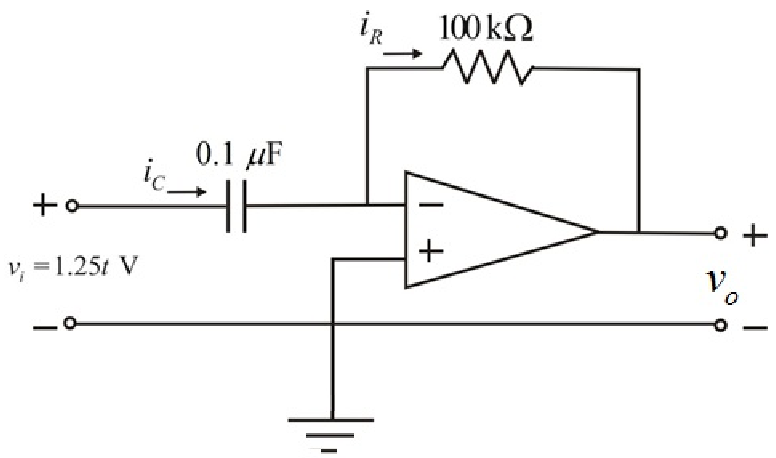



Figure (1)

**Step 2** of 2

The dc voltage, $v_i = 1.25t \text{ V}$

Write the formula for output voltage, $v_o$.

$$v_o = -RC\frac{dv_i}{dt}$$

The current through the resistor is equal to that in the capacitor, that is, $i_R = i_C$ because both resistor and capacitor are in series.

Thus,

$$\begin{aligned} v_o &= -(100\times10^3)(0.1\times10^{-6})\frac{d}{dt}(1.25t) \\ &= (-0.01)(1.25) \\ &= -0.0125 \\ &= -12.5 \text{ mV} \end{aligned}$$

Thus, the output voltage, $v_o$ is $\boxed{-12.5 \text{ mV}}$.